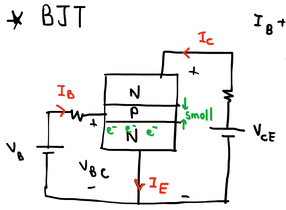
1. Mosfets

07/03/25 9:32 PM

* BJT



$I_B + I_C = I_E$

$I_C = \alpha I_E$

$I_B = I_E(1-\alpha)$

$= I_C\left(\frac{1-\alpha}{\alpha}\right)$ } β

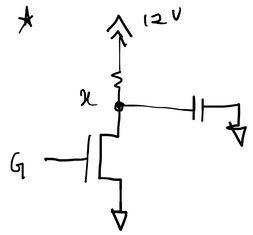
$I_B = \beta I_C$

CCCS

G → S ; D → G

negligible i no i

*

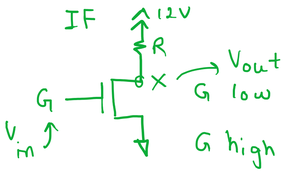


t=0 GS<0 x=0

t=∞ x=12V

if GS>0 capacitor disch. then $V_x = 0$ again

IF

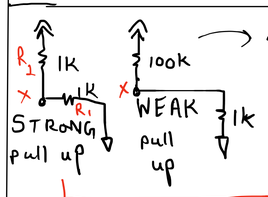


G low X = 12 (pull up)

G high X = 0

V_i	V_o
L	H
H	L

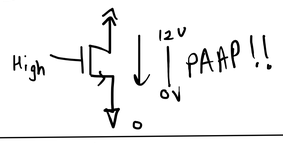
INVERTER



↑ ΔV cuz of higher R

$R_2 << R_1$ toh strong pull ↑

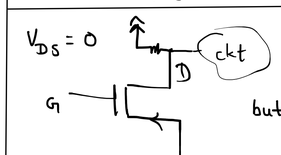
R_2 can't be 0 tho



PAAP!!

kaboom!!

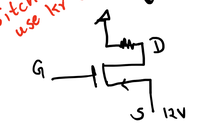
$V_{DS} = 0$



if G low, connected but $V_{DS} = 0$ abhi bhi

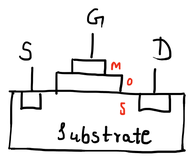
islie top ko pull down KRO

Switch ki tarah use kr skte hai



$V_{DS} < 0$

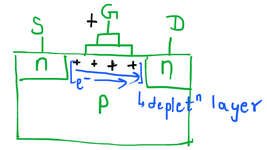
Metal oxide semiconductor field effect transistor –



N mos

P mos

substrate type iska opposite



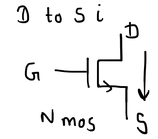
V_{DS} +ve

V_G +ve ; $V_G > V_{TH}$

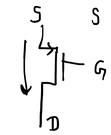
[if $V_G < V_{th}$ toh no i]

Unless V_G crosses thresh-hold V no i will flow thru D to S.

D to S i



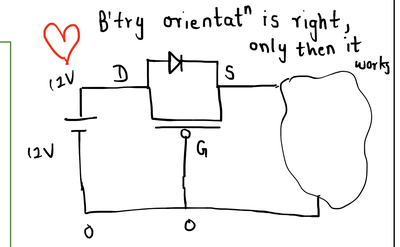
S to D i



Nmos : S → Ground, D → 12V

Pmos : S → 12V, $V_{SD} > 0$, D → ground

B'try orientatⁿ is right, only then it works



Modes :

*Enhancement

channel DNE at first. With $V_{applied}$ channel is created

*Depletion

channel already exists! $V_{applied}$ kinda 'removes' the channel.

Nmos: (ignoring V_{TH})

$V_{GS} > 0$ to closed, conduct krega

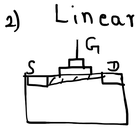
Pmos: $V_{GS} < 0$ conduct ✓

Operating regions- (Nmos)

1) Cutoff $V_{GS} < V_{TH}$

No conductⁿ

2) Linear

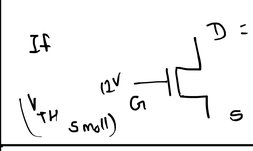


$V_{DS} < V_{GS}$ $I_D = K\left(V_{GS} - \frac{V_{DS}}{2}\right)V_{DS}$

3) Saturatⁿ $I_D = K(V_{GST})^2$

$V_{DS} > V_{GS}$ $V_{GS} - V_{TH}$

If

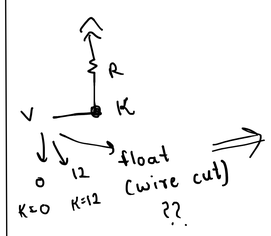


(V_{TH} small)

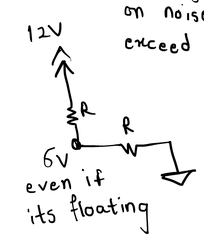
toh $V_S = 12V$ $V_{GS} > V_{Th}$

cuz conduct krna chahiye

High impeadance state –



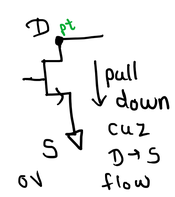
Floating pt may pick up on noise & that may exceed V_{TH}



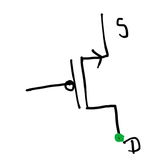
NMos ⇒ $V_{GS} > 0$

(Sat.) $V_{DS} > V_{GS} > 0$

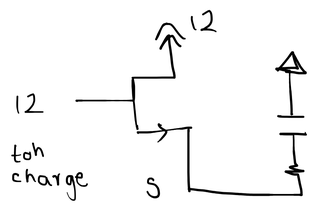
D → S



pull down cuz D→S flow



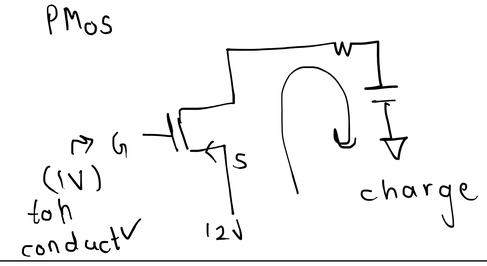
Nmos :

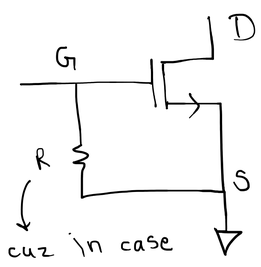


toh charge

OK to NOPE

PMOS





cuz in case G becomes floating toh woh pull down ho.

NO I thru GATE!!